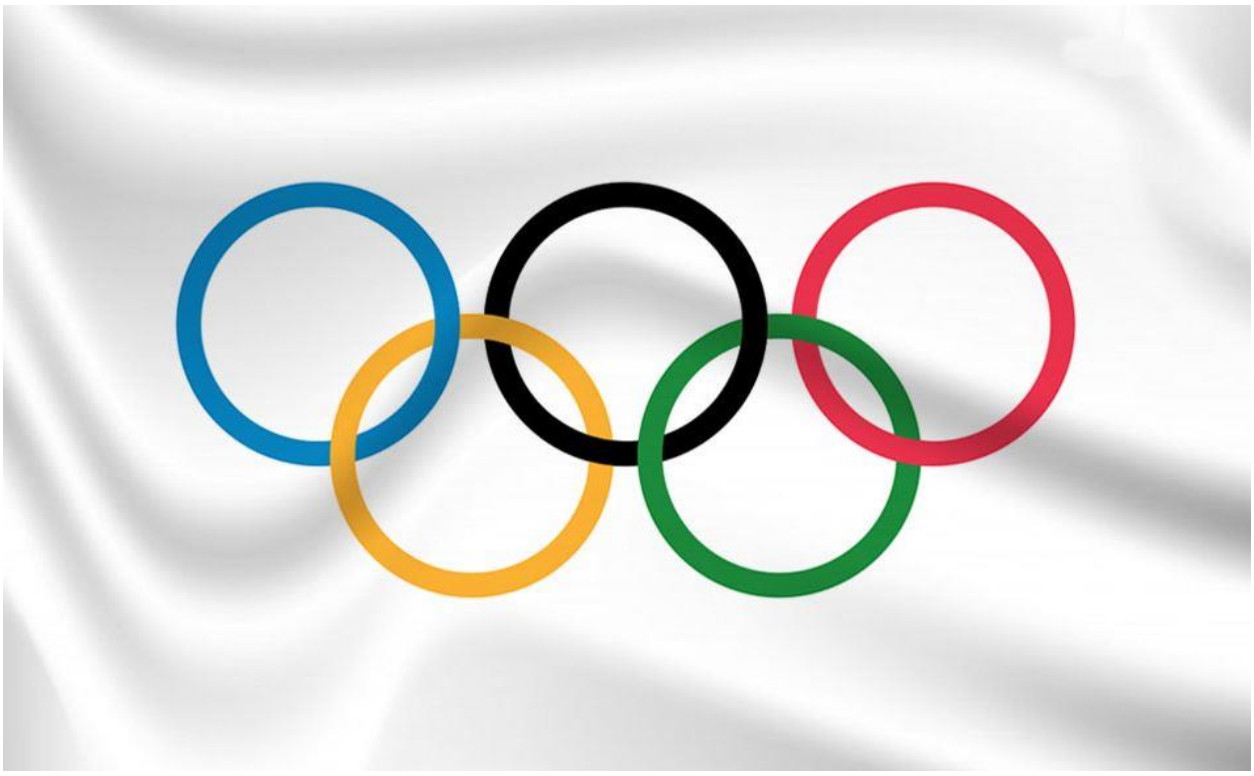
SQL FOR Data Science

Assignment – I

Comprehensive Analysis of Olympic Medals By Country (1896-2020)



Team members :

D/ADC/24/0005

D/ADC/24/0023

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Power BI, developed by Microsoft, is a powerful tool for visualizing and analyzing data. It allows users to create interactive dashboards and reports, connecting to various data sources like databases and spreadsheets. With features like real-time analytics and data modeling, Power BI helps users explore data dynamically and supports advanced analysis. It also enables easy sharing of reports and dashboards within teams or organizations, making it essential for informed decision-making. By using power BI dashboard we create this report for presents a comprehensive analysis of Olympic Games data from 1896 to 2020

Summary

This report presents a comprehensive analysis of Olympic Games data from 1896 to 2020 using Power BI to visualize and interpret key metrics, trends, and insights. The primary focus is on the performance of various countries in both Summer and Winter Olympics, highlighting medal counts, trends over time, and the distribution of gold, silver, and bronze medals. The main key points are as follows:

- Total Medals Analysis : A total of 156 teams competed in the Olympics, winning 20,281 medals.
- Summer vs. Winter Olympics: The dashboard showcases the total medals won by each country in the Summer vs. Winter Olympics.
- Medal Count Comparison: The United States, Soviet Union/Russia, and Great Britain are the top three nations in terms of total medals. France and Germany have comparable medal totals, with equal contributions from Summer and Winter Games. China shows a recent surge in Winter Olympic medals.
- Trends Over Time: The dashboard reveals consistent medal-winning performances by the United States. China has seen a significant rise in medals, particularly in Winter Olympics in recent years.
- Medal Distribution by Type: The distribution of gold, silver, and bronze medals for the top nations shows varying percentages, with China and the United States having a high percentage of gold medals.
- Participation vs. Success Rate: The analysis indicates that nations like the United States, with extensive participation, have higher success rates. China, despite fewer participants, has shown noteworthy success, especially in recent years. In conclusion,

This thorough examination of Olympic medals from 1896 to 2020 highlights the dominance of the United States, Soviet Union/Russia, and Great Britain in both Summer and Winter Games. Recent trends indicate a rise in performance from countries like China, particularly in Winter sports. The distribution of gold, silver, and bronze medals underscores the competitive spirit and excellence of these nations. Overall, this report provides a detailed overview of the global Olympic success landscape, showcasing historical dominance and emerging trends in the world of sports.

Introduction

The Olympic Games, held every four years, are the pinnacle of global sports competition. Since their modern inception in 1896, these games have provided a platform for athletes worldwide to showcase their talents in both Summer and Winter events. This report delves into a comprehensive analysis of Olympic medals won by various countries, comparing their performances in Summer and Winter Olympics from 1896 to 2020.

The objective of this assignment is to create a Power BI dashboard that effectively visualize data from previous Summer and Winter Olympic Games. This dashboard will serve as a comprehensive tool to analyze and interpret various aspects of the Olympic data, providing insights into trends, performances, and other key metrics from past events.

Methodology

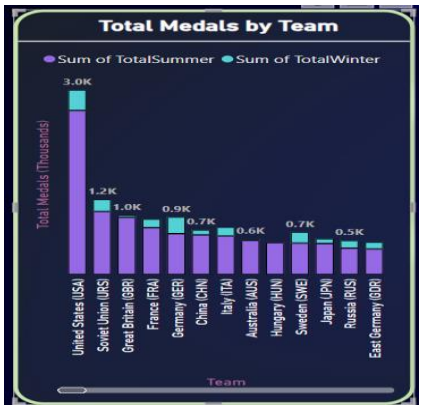
We collected Data from Wikipedia page (All-time Olympic Games medal table.) The needed dataset can be found at the following location:

(https://en.wikipedia.org/wiki/All-time_Olympic_Games_medal_table)

Extract the tables containing the all-time medal counts for both Summer and Winter Olympics collection data with following columns: Country, Number of Summer/Winter Games participated, Gold medals, Silver medals, Bronze medals, Total medals. Then import the data into a excel for cleaning and transformation. Next, we imported the copied dataset in to the power BI after cleaning the data and handle any missing values, inconsistencies, or formatting issues. Before imported to the power BI ensure that the data is structured in a way that Power BI can easily import and use it. Then we Created a Power BI dashboard that effectively visualize the following data analyzing parts.

Total medals won by each country in Summer vs. Winter Olympics.

The analysis includes a total of 156 teams that have participated in the Olympics, collectively winning 20,281 medals. The dashboard provides a detailed breakdown of these medals, highlighting the top-performing countries. Purple color represents the sum of total medal won by each country in summer and blue color represents the sum of total medal won by each country in winter. Relative to the others countries USA won highest number of medals in summer & winter.



Comparison of medal counts across different countries.

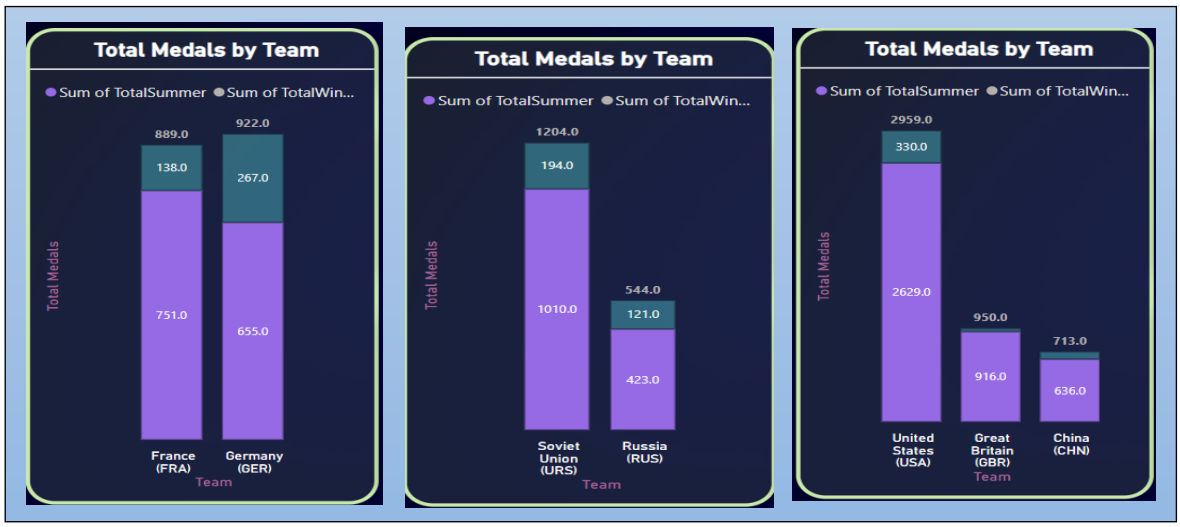
- United States: With 2959 medals overall, the United States is in first place. Both the Summer (2,629) and Winter (330) games have contributed significantly to this total.

- Soviet Union/Russia: With impressive results of Soviet Union both the Summer (1010) and Winter (194) events, the Soviet Union, including its successor Russia, has amassed around 2,200 medals.

- Great Britain: With 950 medals, this country has performed admirably, especially in the Summer Games (where it won 916 medals) winter games (where it won 34 medals).

- France and Germany: With 1811 medals won respectively,751 medals have been won in France summer.138 have been won in winter.655 medals have been taken in Germany summer. 267 have been won in winter.

- China: China has won 713 medals in total, primarily in the Summer Games (636) and an increasing amount in the Winter Games (77), thanks to its recent surge in Olympic sports.



Trends over time for selected countries or overall.

The dashboard shows that while some nations, like the United States, have routinely won the most medals at both the Summer and Winter Olympics, other nations, like China, have witnessed a substantial rise in medal totals recently, especially in the Winter Olympics.

Distribution of gold, silver, and bronze medals

The "Medal Distribution by Type" section shows the proportion of gold, silver, and bronze medals for the top countries:

- *United States*: 39% gold, 32% silver, 28% bronze

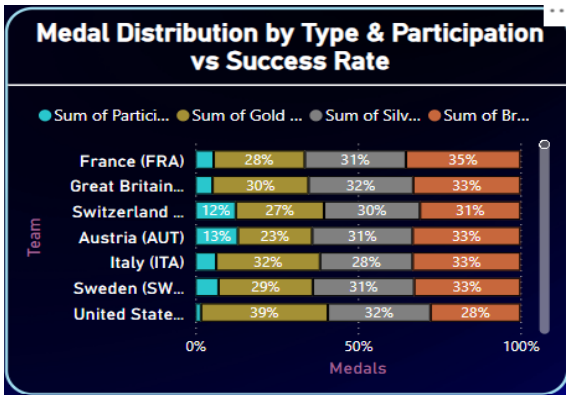
- *Italy*: 32% gold, 28% silver, 33% bronze

- *Australia*: 23% gold, 31% silver, 33% bronze

- *France*: 28% gold, 31% silver,35 % bronze

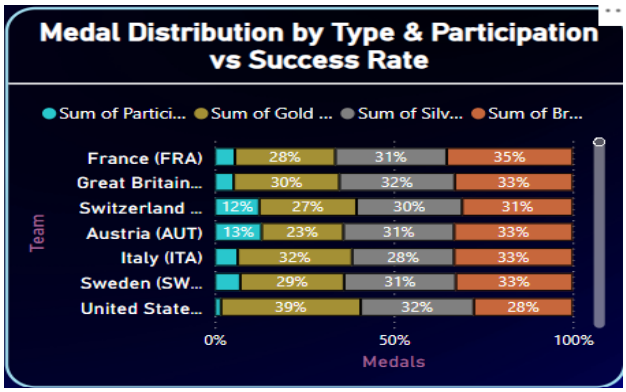
- *Sweden*: 29% gold, 31% silver, 33% bronze

This distribution highlights that some countries not only win a high number of medals but also a significant proportion of them are gold.



Participation vs. success rate (number of games participated vs. medals won).

The participation versus success rate can be inferred from the number of games participated in and the medals won. Countries like the United States, with extensive participation in almost every edition of the games, have a higher success rate. On the other hand, countries like Italy have fewer participations but have shown a remarkable success rate, especially in recent years. Blue color represents the number of participants rate and the others as gold, silver, bronze respectively.



Final Thoughts

The study examines Olympic medals from 1896 to 2020, revealing the US, UK, and USSR/Russia as the top three nations, with China's performance increasing in winter sports. The paper provides a comprehensive overview of global Olympic success.